Calculus 1m - 104018 - Spring Semester

Final Exam A

Instructions:

You must answer on these pages. You may not add pages or papers to this exam.
You must write with a black or a blue pen. Do not use a pencil or a red pen.

You may not use notes, books, calculators or electronic devices of any kind.

Good Luck!

Part A

Mark the correct answer in each of the questions. Each of the first 4 questions is 8 points.

1. Let $f(x) = \begin{cases} \sin\frac{1}{x} & x \neq 0 \\ -\frac{1}{3} & x = 0 \end{cases}$. Mark the correct answer.

   (a) $\int\limits_{-1}^{1} f(x)dx = 0$.

   (b) The function is one to one (injective) on $(0, \infty)$.

   (c) $\int\limits_{0}^{1} f(x)dx > 1$.

   (d) $f(x)$ is increasing on $(-\infty, 0)$.

   (e) $f'(x)$ is bounded on the interval $(0, \infty)$.

2. Consider the improper integrals

$$I = \int_{1}^{\infty} x \sin\left(\frac{1}{x^{\frac{1}{2}}}\right) \arctan\left(\frac{1}{x}\right) dx \quad II = \int_{0}^{1} \frac{\ln x}{\sin\left(\sqrt{x}\right)} dx.$$

(a) I does not converge and II converges.

(b) Both improper integrals converge.

(c) I converges and II does not converge.

(d) None of the improper integrals converges.

3. Consider the series

$$I = \sum_{n=1}^{\infty} n^2 \left(1 - \cos\left(\frac{1}{n^2}\right)\right) \qquad II = \sum_{n=1}^{\infty} \frac{(n+1)^n}{2^n \cdot n!}$$

(a) Both series converge.

(b) Both series do not converge.

(c) Series I does not converge and series II converges.

(d) Series I converges and series II does not converge.

4. Let $f(x)$ be defined for all x. Mark the correct statement.

(a) If $f(x)$ is not continuous at 0, then $\left(f(x)\right)^2$ is not differentiable at 0.

(b) If $f'_-(x)$, $f'_+(x)$ exist for all x, then $f(x)$ is continuous at every point.

(c) If $f(x)$ is convex on the interval $(-1, 1)$ then $f(x)$ is differentiable on the interval $(-1, 1)$.

(d) If $f''(x) > 0$ for all x, then $f(x)$ has a unique minimum point.

5. (12 points) Answer each of the following True or False questions:

(a) If $\lim\limits_{n\to\infty} a_n$ exists and isn't zero, and $\lim\limits_{n\to\infty} a_n \cdot b_n$ exists, then $\lim\limits_{n\to\infty} b_n$ exists.

True False

(b) If $\sum\limits_{n=1}^{\infty} a_n^3$ converges, then $\sum\limits_{n=1}^{\infty} a_n^6$ converges.

True False

(c) If $\sum\limits_{n=1}^{\infty} a_n$ converges absolutely, then $\sum\limits_{n=1}^{\infty} \sin(a_n)$ converges.

True False

(d) If $a_n \xrightarrow[n\to\infty]{} 0^+$ (meaning the sequence is positive and tends to zero), then $\ln\big(1+a_n\big) \le a_n^2$ for all $n \ge 1$.

True False

Part B

Answer each of the following questions fully. Explain all your answers.

6. (28 points)

(a) (5 points) State Rolle's theorem.

(b) (12 points) Prove Rolle's theorem.

(c) (11 points) Let $f(x)$ be a differentiable function on the interval $[a, b]$ for which $f'_+(a) > 0,\ f'_-(b) < 0$. Prove that there is $a < c < b$ for which $f'(c) = 0$.

7.(a) (9 points) Calculate

$$\int \frac{6x+7}{x^2+4x+7}dx$$

(b) (10 points) Calculate $\sqrt{1.1} = (1.1)^{\frac{1}{2}}$ with accuracy 10^{-4}.

(c) (9 points) Calculate

$$\lim_{x \to 0} \frac{\int_{x^2}^{x^3} \sin(t^2) dt}{\sin x^6}$$